RAJAGANAPATHY M

AI / GenAI Engineer

LLM and Agentic AI · RAG · Computer Vision · Full-Stack AI · Cloud Deployment

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Summary

3+ years in AI/ML engineering (since 2023), M.Tech in AI (CGPA 9.6/10), immediate joiner. AI engineer who builds full-stack AI products: model, API, UI, cloud deployment and monitoring. Shipped a bilingual medical AI assistant to a paying client, built a multi-agent coding system for a startup, and authored a patent-filed, IEEE-published reasoning framework (+250% FEVER F1 over chain-of-thought).

Technical Skills

GenAI / LLMs	RAG (FAISS, Qdrant), LangChain, HuggingFace, QLoRA/LoRA/PEFT, Groq, Gemini, GPT-4o, LLM orchestration
Agentic AI	Google ADK, multi-agent systems, tool calling, LLM-as-Judge, agentic workflows
DL & Vision	PyTorch, TensorFlow, CNN, LSTM, Transformers, CLIP, BLIP, ViT, OpenCV, OCR, VQA
Backend & Frontend	Python, FastAPI, REST APIs, JWT, Firebase Auth, React, JavaScript, Streamlit, Gradio
Cloud & MLOps	Azure (Container Apps, OpenAI), AWS (SageMaker, S3, EC2), GCP (Vertex AI), Docker, GitHub Actions, Terraform, Grafana

Projects

Anbu Health AI (live, paid client) *React, FastAPI, Azure, Qdrant RAG*

- Bilingual (Tamil/English) assistant that analyses lab reports, scans and medicine photos; live at anbuclinic.me and used by real village patients.
- 5-stage pipeline: routing → RAG → LLM (Groq, GPT-4o vision) → confidence scoring → hallucination check; deployed with Terraform, GitHub Actions and Prometheus/Grafana.

Spec-Driven Multi-Agent Coding System (startup client, in development) *Google ADK, Gemini 2.5 Flash*

- 3-level agent hierarchy (Orchestrator → Feature Leads → Explorer, Coder, Tester, Reviewer) with sandboxed file tools, a git checkpoint per task, and golden-task plus LLM-as-Judge evaluation; delivered a HIPAA-aligned healthcare QA app.

Antahkarana Reasoning Framework *Qwen2.5-7B, BLIP-2*

- 5-module reasoning architecture for LLMs and VLMs: **+59.61% TruthfulQA**, **+250.82% FEVER F1** over chain-of-thought, **26% fewer model calls** than Self-Consistency, VQA hallucination 26.6% → 20.9%; powers Anbu Health AI.

MML Smart Campus Security *CLIP, BLIP, OpenCV*

- CLIP threat classification, BLIP visual Q&A and voice biometrics to replace manual CCTV monitoring.

CodeLlama-7B Coding Assistant *AWS SageMaker, QLoRA, HF Spaces*

- QLoRA fine-tune on CodeAlpaca-20K, **cutting GPU memory 60%**; live Gradio demo on HuggingFace Spaces.

Experience

Founder & Independent AI/ML Engineer Jul 2023 – Present
AI Vision (MSME-registered), Chennai

- Own the full product lifecycle for client AI work: frontend, backend, AI pipeline, cloud infrastructure and monitoring.
- Building a multi-agent coding platform for a startup client; 3 models published on HuggingFace.

Electrical Construction Site Engineer Jan 2017 – Jun 2023
SR Electrical Works, Chennai

- Sole engineer on 6 projects (2 hospitals, 2 hotels, 2 colleges; Rs. 5–6 Cr, team of 20) with **zero site rejections**; grew profit **10**× by restructuring procurement and billing (2017).
- Applied AI (2021–2023): OCR and CAD automation for symbol detection and quantity take-offs from drawings.

Electrical Maintenance Engineer Dec 2014 – Dec 2016
Mod Forge Pvt Ltd, Chennai (ISO/TS 16949)

- Maintained 500 kVA supply, 250 kVA DG set and motors up to 400 kW on CNC/VMC lines with zero unplanned downtime for 2 years.

Education

M.Tech, Artificial Intelligence (CGPA 9.6/10), SRM Institute of Science & Technology — 2024 – 2026
Certified AI & ML Program, IIT Madras Pravartak (GUVI, NASSCOM) — Nov 2023 – Nov 2024
B.E., Electrical & Electronics Engineering, Thangavelu Engineering College, Chennai — 2013

Patent, Publication & Certifications

- **Indian Patent Application** 202641043947 (Apr 2026)
- **IEEE Xplore paper**, Antahkarana framework
- **AWS Solutions Architect Associate** (Oct 2025)
- **Kaggle AI Agents Intensive** with Google (Dec 2025)